

Teacher Guide — primary / module-14- magic-squares

IKS Curriculum

Table of Contents

Overview

Module 14 — Magic Squares (Primary)

Age band: Primary · **Grades:** 3–5 · **Duration:** 10 sessions × ~30 min

What this module is

Children meet the **3×3 magic square** as a puzzle: place the numbers 1 to 9 in a grid so that every row, every column, and the two diagonals all add up to the same number. They discover by play that the magic number is **15** and that the same arrangement keeps coming up (just rotated). Along the way they hear the story of the **Lo Shu turtle** (China) and meet a stone carving from **Khajuraho** (India) that looks like the same idea.

What the children will be able to do

By the end of this module: - Recognise a 3×3 magic square when they see one. - Build a 3×3 magic square with 1–9 either from memory or with a hint. - Add three single-digit numbers and check whether the sum is 15. - Tell, in two or three sentences, where magic squares come from in stories (Lo Shu turtle, Khajuraho temple stone). - Sing the “magic square song” — “*All to fifteen, all to fifteen*”.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems in age-appropriate, story-led form.
- **§4.23** — Experiential learning through hands-on tile placement and discovery.
- **§4.27** — Local context: the Khajuraho temple stone is a real Indian artefact children can locate on a map.

What we won’t do at this band

- No formula ($n(n^2 + 1)/2$ is Middle-band material).
- No 4×4 or 5×5 construction.
- No formal Sanskrit citations to students. Sources used only by teacher.
- No ritual or religious framing of the diagrams.

Materials checklist

- A class set of paper tiles or wooden tokens numbered 1–9 (1 set per pair)
- Printed blank 3×3 grids (~5 per child across the module)
- Coloured pencils (4 colours per child)
- A simple picture of a turtle for Day 7 (for the Lo Shu story)
- A picture or photocopy of the Khajuraho stone-square panel for Day 8
- Big chart paper for the class wall: “OUR MAGIC SQUARE — sum 15”

Day-by-day overview

Day	Title	Focus
1	The Magic Number 15	Hook: warm-up addition game
2	Pairs that add to 10	Foundation: making addition fluent
3	The First Try	Children try the 3×3 by hand
4	The Secret of the Centre	Always 5 in the middle
5	Our Class Magic Square	Build it together on a big chart
6	Magic Square Sing-along	A chant + tile game
7	The Lo Shu Turtle	Story from China
8	The Khajuraho Stone	A magic square from India
9	Make-and-Keep	Each child decorates their own magic square
10	Show and Tell	Children show their magic square + sticker assessment

Project (light)

Each child colours and decorates a magic square page they have built themselves. Stickers awarded based on: - Sum-check stickers (verified each row, column, diagonal) - Story-telling stickers (knows the Lo Shu story OR the Khajuraho story) - Effort sticker - Helping-a-friend sticker

Sources used (teacher only)

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE (primary source — for the construction method shown to children).
- Plofker (2009), *Mathematics in India* (secondary teacher reference).

Children do not see these citations directly at Primary band.

10-Day Lesson Plan

Module 14 (Primary) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / artefact
1	The Magic Number 15	days/day-01-magic-number.md	Wall chart “to 15”
2	Pairs that add to 10	days/day-02-pairs-to-ten.md	Tile pair game
3	The First Try	days/day-03-first-try.md	Worksheet: blank 3×3
4	The Secret of the Centre	days/day-04-secret-centre.md	Class discovery
5	Our Class Magic Square	days/day-05-class-square.md	Big chart on wall
6	Magic Square Sing-along	days/day-06-singalong.md	activities/activity-01-tile-game.md
7	The Lo Shu Turtle	days/day-07-lo-shu-turtle.md	Story + draw a turtle
8	The Khajuraho Stone	days/day-08-khajuraho-stone.md	India map find
9	Make-and-Keep	days/day-09-make-and-keep.md	Decorate own square
10	Show and Tell	days/day-10-show-and-tell.md	Sticker assessment

Pacing notes

- Sessions are ~30 minutes. Be flexible — Days 3 and 4 can each stretch if children are still trying combinations.
- The chant (“all to fifteen, all to fifteen”) is sung at the start of every class from Day 2 onwards. Repetition is the lesson.
- Days 1–5 are the math; Days 6–8 add story and culture; Days 9–10 are the project.

Daily chant

“One, two, three — count to fifteen! Four, five, six — still fifteen! Seven, eight, nine — all to fifteen!”

(Use during settle-in. Clap on each number.)

What children leave with

- A coloured 3×3 magic square they made themselves.
- The chant “all to fifteen.”
- Two stories: the **Lo Shu turtle** and the **Khajuraho stone**.
- Confidence adding three single-digit numbers.

Teacher Notes

Teacher Notes — Module 14 Magic Squares (Primary)

Posture

At Primary band, **the puzzle is the lesson**. Children discover that putting 1 to 9 into a 3×3 grid so every row, column, and diagonal sum to 15 *is hard* — and then discover that there’s a satisfying pattern (5 in the centre, even numbers at the corners). They do not learn the formula and they do not construct anything bigger than 3×3 .

The cultural content (Lo Shu turtle, Khajuraho stone) is a story-anchor for the math. Use story-time framing — not lecture.

Common misconceptions

1. **“Any nine numbers will work.”** No — we use exactly 1, 2, 3, 4, 5, 6, 7, 8, 9, each once. Reinforce by laying out the tiles.
2. **“There’s only one row to check.”** Make sure children count 3 rows + 3 columns + 2 diagonals = 8 sums to check. Eight stickers on a verified square.
3. **“The middle can be any number.”** Quietly: 5 must be in the middle. We show this on Day 4 by hand (not algebra).
4. **“Magic squares are magic.”** Reframe: “We call them magic squares because every line adds up to the same — that *feels* magic, but it’s just careful arrangement.”

Differentiation hints

- **Tier up:** A child who has built the 3×3 confidently can try to make a 4×4 with 1–16 summing to 34. (They will fail without a method — that’s fine. The struggle is the gift to Middle band.)
- **Tier down:** Stay on the chant and the “sums to 10” pair game. A child who can confidently say “ $8+2=10$, $7+3=10$, ...” is on track even if they haven’t built the full square.

Sensitivity notes

- Some children may have seen magic squares used in religious or astrological contexts at home (e.g. as *yantras*). If a child mentions this, acknowledge warmly — “yes, magic squares are used in many ways. In this class we look at the numbers.” Do not introduce ritual framing yourself.

- No timed competition between children. Praise effort and persistence, not speed.
- Children with arithmetic anxiety: pair them with a confident partner. The tile-laying is physical and helps.

Behind the scenes (don't say to children)

The 3×3 magic square is *unique up to symmetry* — there are 8 arrangements (rotations + reflections) of the same underlying square. Children will often produce mirror images and think they're different. They are — and they aren't. At Primary band: praise both and don't worry about it.

Project oversight (the Make-and-Keep)

- Day 9 is the make-and-decorate session. Most children will need help making sure their square is actually magic (i.e. that they didn't write 4 twice). Walk around and check.
- Day 10's "show and tell" is gentle. Each child speaks for 30 seconds about their square. Family-style audience.

Safety

- Paper tiles only (no small parts that can be swallowed for younger Primary children).
- Coloured pencils only for decoration; no markers that bleed through.

If you only have 6 days

- Day 1 (Magic Number 15)
- Day 2 (Pairs to 10)
- Day 3 (First Try + Secret of the Centre — combined)
- Day 4 (Lo Shu Turtle — story)
- Day 5 (Make-and-Keep)
- Day 6 (Show and Tell + chant)

(Cut Khajuraho, sing-along, and the class chart.)

Sources

Sources — Module 14 Magic Squares (Primary)

For the teacher (not shown to children at this band)

Primary historical source

Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE.

- The systematic Sanskrit treatment of magic squares we draw on indirectly. At Primary band the children meet the *result* (the 3×3 with 1–9 sums to 15) without meeting the systematic method.

Secondary scholarly source

Kim Plofker, *Mathematics in India* (Princeton University Press, 2009).

- Background reading. Used for historical context only.

Material artefacts referenced

- **Lo Shu (China).** Earliest documented in *Da Dai Liji* (Han dynasty, ~5th–1st c. BCE). Legend of Emperor Yu and the turtle from the Lo river: a *story*, not historical fact. We tell the story as a story.
- **Khajuraho 4×4 inscription (Madhya Pradesh, India).** Chandela period, ~10th–11th c. CE. We show the artefact and locate it on a map of India. At Primary band we acknowledge the existence; we do not work the 4×4.

What to tell children about sources

At Primary band: nothing in formal citation form. The Khajuraho stone is “a real carving in a real temple in India” (not “an inscription dated c. 10th c. CE”). The Lo Shu is “a story from China about a turtle” (not “Da Dai Liji, Han dynasty”).

The framing matters: stories are stories; carvings are real things; numbers are numbers. Keep the language age-appropriate without losing accuracy.

Citations standard for this band

Per the style guide: **Primary band has no student-facing citations.** Citations belong in this teacher-facing sources document only.

Day 1 — Discover the 3×3

Module: Magic Squares — Bhadraganita (Primary) · **Time:** 30 min

Goal

Build the unique 3×3 magic square with 1–9 by trial.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)
- BIG puzzle! Put the numbers 1 to 9 in a 3×3 grid. Each row, column, and diagonal should add to the SAME number!

Children work in pairs. Try, fail, try again.

Hint after 5 min: “*5 always goes in the middle.*”

The answer: 2-7-6 / 9-5-1 / 4-3-8. All rows = 15! All columns = 15! Even diagonals = 15! MAGIC!

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 2 — The Magic Constant

Module: Magic Squares — Bhadraganita (Primary) · **Time:** 30 min

Goal

Derive why 3×3 with 1–9 must sum to 15 per row.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

Today's idea (10 min)

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. "the common sum of all rows/columns/diagonals")
- The MAGIC NUMBER for 3×3 is 15. Why 15?

Add all the numbers from 1 to 9: $1+2+3+4+5+6+7+8+9 = 45$.

Spread 45 across 3 rows: $45 / 3 = 15$. So each row must equal 15!

For a 4×4 magic square (using 1 to 16), what's the magic number? - $1+2+\dots+16 = 136$. / 4 rows = **34**.

"There's a pattern!"

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 3 — How Many 3×3 Magic Squares?

Module: Magic Squares — Bhadrageṇita (Primary) · **Time:** 30 min

Goal

Count the equivalence classes (up to symmetry).

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa's algorithm for odd-order squares”)
- How many ways can we make a 3×3 magic square?

Show the 2-7-6 / 9-5-1 / 4-3-8 square. Rotate the paper 90° — STILL magic! Rotate again — still! Reflect (mirror) — still!

“*That's 8 ways! But they're all REALLY the same square — just turned around.*”

Children try rotating their own magic square paper. They count the rotations.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 4 — Stair-Step Method (Nārāyaṇa)

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Apply Nārāyaṇa's algorithm to build a 5×5 square.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. "auspicious arithmetic" — the Indian term for magic squares)
- Big-kid trick! Nārāyaṇa (an Indian mathematician 700 years ago) taught how to BUILD a magic square step-by-step.

Show on the board: 1. Put **1** in the top-middle. 2. From 1, jump UP-and-RIGHT to put 2... but you fall off the top! 3. Wrap to the bottom row (same column). 4. Keep going!

The pattern looks like a horse jumping. *"That's why he called it the horse-step!"*

Children watch the teacher build a 3×3 step-by-step.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 5 — Practice — 5×5 and 7×7

Module: Magic Squares — Bhadraganita (Primary) · **Time:** 30 min

Goal

Use the method on multiple odd-order squares.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. “the common sum of all rows/columns/diagonals”)
- Today: try a BIGGER magic square — 5×5 !

Place 1 in the top-middle. Step up-right. Wrap when needed. Keep going to 25.

It looks confusing at first, but the rule is the same. The magic number for 5×5 is **65** (all rows, columns, diagonals add to 65).

Children try with the teacher's help. Most won't finish — that's OK! Just enjoy the start.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 6 — 4×4 and the Khajuraho Square

Module: Magic Squares — Bhadrageṇita (Primary) · **Time:** 30 min

Goal

Examine the Khajuraho inscription and its panel-magic property.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa's algorithm for odd-order squares”)
- 4×4 magic squares are also amazing! In INDIA, there's one carved on a temple wall — the Khajuraho square. About 1000 years old!

Show the picture. “*All rows = 34, all columns = 34. Even small 2x2 boxes inside = 34!*”

Children draw a 4×4 grid and copy the numbers. They check: do the rows add to 34? Try one!

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — Historical Context — Nārāyaṇa and Predecessors

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Place *Gaṇita-kaumudī* in 14th c. Indian math.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)
- Nārāyaṇa Paṇḍita lived in India about 700 years ago. He wrote a book about magic squares!

His book is called *Gaṇita-kaumudī* (“Moonlight of Math”). Book 14 is all about magic squares.

Show a (drawn) picture of Nārāyaṇa with a manuscript. “*Maths heroes from long ago!*”

Children draw a magic square inside a “book” outline.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Cross-Cultural — Lo Shu and Dürer

Module: Magic Squares — Bhadrageṇita (Primary) · **Time:** 30 min

Goal

Compare Indian, Chinese (Lo Shu), and European (Dürer 1514) magic squares.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *siddhāṅka* (*modern coinage*) (IAST: *siddhāṅka* (modern coinage); lit. “the common sum of all rows/columns/diagonals”)
- Magic squares are found ALL OVER THE WORLD!

-  China: Lo Shu legend — a turtle’s shell!
-  India: Khajuraho temple wall!
-  Germany: Dürer’s painting!

Show 4 small pictures. “*People everywhere love magic squares!*”

Each child picks one tradition and draws a tiny picture (turtle, temple, painting).

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Design Your Own

Module: Magic Squares — Bhadraganita (Primary) · **Time:** 30 min

Goal

Create a personalised 4×4 or 5×5 magic square as visual art.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today's idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa’s algorithm for odd-order squares”)
- Make YOUR OWN magic square as art!

Choose ANY 9 numbers (or use 1-9). Make a 3×3 grid where all rows add to the same. Decorate it!

Show example: a magic square shaped like a flower, with colours on each number. “*Math + art = magic!*”

Children draw, colour, and write their magic number at the top.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Share and Assessment

Module: Magic Squares — Bhadraganita (Primary) · **Time:** 30 min

Goal

Present design + summative quiz.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “What’s one thing you remember from yesterday?”

Today’s idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)
- Final day! Show your magic square art.

Each child shows + says one thing they liked. Cheer for each.

Quiz with stickers — everyone gets a “Magic Squares” sticker.

End: chant — “*Bhadragāṇita! Magic numbers!*”

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Formative Quiz — Magic Squares — Bhadragaṇita (Primary)

When: mid-module (after Day 5). **Duration:** 20 minutes. **Format:** mix of short answer, multiple choice, and reflection. Sticker self-assessment for Primary.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Magic Squares — Bhadragaṇita in one sentence.
3. (Multiple choice) Which day in this module focused on ‘How Many 3×3 Magic Squares?’? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4

4. True / False — and explain in one sentence: *Magic Squares — Bhadraganita is purely a religious concept and has no scientific framing.*
5. Connect Magic Squares — Bhadraganita to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘bhadrā-gaṇita’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about Magic Squares — Bhadraganita and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it’s a check-in to see what’s landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Magic Squares — Bhadraganita (Primary)

When: end of module (after Day 10). **Duration:** 30 minutes. **Total points:** 20.

Questions

Part A — Vocabulary (~5 points) A1. Define *bhadra-gaṇita* in your own words. A2. Define *siddhāṅka* (*modern coinage*) in your own words. A3. Define *kothumi vidhi* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Magic Squares — Bhadraganita in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Magic Squares — Bhadraganita with a concept from another module (your choice). What’s similar, what’s different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Magic Squares — Bhadraganita from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What’s one question you still have?

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.

- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 12/20 indicates the student has met the module's learning objectives.

Homework

Homework — Magic Squares — Bhadragaṇita (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — Discover the 3×3** · Draw about today's concept. (~5 min)
2. **Day 2 — The Magic Constant** · Draw about today's concept. (~5 min)
3. **Day 3 — How Many 3×3 Magic Squares?** · Draw about today's concept. (~5 min)
4. **Day 4 — Stair-Step Method (Nārāyaṇa)** · Draw about today's concept. (~5 min)
5. **Day 5 — Practice — 5×5 and 7×7** · Draw about today's concept. (~5 min)
6. **Day 6 — 4×4 and the Khajuraho Square** · Draw about today's concept. (~5 min)
7. **Day 7 — Historical Context — Nārāyaṇa and Predecessors** · Draw about today's concept. (~5 min)
8. **Day 8 — Cross-Cultural — Lo Shu and Dürer** · Draw about today's concept. (~5 min)
9. **Day 9 — Design Your Own** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Share and Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Activity 1 — Class 3×3 Magic-Square Tournament

Module: Magic Squares — Bhadragaṇita · **Band:** Primary · **Duration:** 25 min

Purpose

Reinforce the day-by-day concepts of Magic Squares — Bhadragaṇita through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

grids on paper, pencils, timer

What students do

In pairs, students race to construct the 3×3 magic square with 1–9 from a starting 9-position permutation. First pair with correct sums wins.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (15 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Magic Square as Visual Art

Module: Magic Squares — Bhadraganita · **Band:** Primary

What you will do

Design a 4×4 or 5×5 magic square. Decorate it as a piece of visual art (mandala-style, calligraphic, geometric). Include the magic constant calculation on the back.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.

- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a drawing / made object).
- **Written reflection** (oral share): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (3-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 1 sources. Use the form: *Author, Title, year.* (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Magic Squares — Bhadragaṇita (Primary)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

- **Knows the new word(s)** — 🌟 Strong understanding / 🌱 Growing / ☁ Needs more practice
- **Joined in activities** — 🌟 Strong understanding / 🌱 Growing / ☁ Needs more practice

- **Asked good questions** — 🌟 Strong understanding / 🌱 Growing / 🌧 Needs more practice
- **Drew / made something** — 🌟 Strong understanding / 🌱 Growing / 🌧 Needs more practice

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.